

EBIS v227 — Annotated Forensic Audit Trail

Runtime session annotation · boot sig 25e76ee6 · evidence verification → drivers investigation

Engine Behavior Intelligence System · Theta combustion kernel · DIFC / ORCH / PWE / SBL / ACIS / CEH · 41 raw log blocks, fully preserved

Method. Every raw log block below is reproduced verbatim and unmodified — no line removed, reordered, summarized, or reformatted. Each block is followed by an annotation explaining what happened, what it means, what evidence it produced, why it matters, and the insight obtained. Annotations are interpretation only; they add nothing to and remove nothing from the original runtime output.

RAW LOG BLOCK — #0

```
Extracting ebis_v22_complete_v227.zip ...

EBIS v22 · boot verified · sig 25e76ee6
47 files · 7/7 modules · 12 templates · manual mode · 2.9s

All modules imported successfully

EBIS · waiting for first investigation
engine offline — 'engine chalao' se shuru karo
→ Investigation context restored: branch=evidence verification last=ca50 vars=6

ENGINE VIEWPORTStatus: STOPPEDCycle : 1655RPM : 2000AFR : 14.7CA50 : +1.2°Spark : 26.9°Knock : HIGH0sc : 1.00
```

EXPLANATION

What happened: EBIS v227 archive extracted and the runtime self-verified at boot: signature **25e76ee6**, 47 files, **7/7 modules** imported, 12 templates, manual mode, 2.9s. Engine starts offline, and a prior investigation context is restored — branch *evidence verification*, last target *ca50*, 6 variables.

What this means: The system integrity-checks itself (signature + module count) before accepting any command, then rehydrates its scientific memory instead of booting blank. The earlier line of inquiry is continued, not discarded.

Evidence produced: Boot signature, module-load ratio (7/7), template count, and the three restored context fields (branch / last target / var count).

Why this matters: A signed, reproducible boot plus restored persistent memory is the precondition for a trustworthy forensic run — it proves continuity across sessions and that nothing partially loaded.

Insights: The lab 'remembers' where it stopped (a *ca50* investigation), so everything that follows builds on existing state rather than starting cold.

RAW LOG BLOCK — #1

```
[ROUTER] COMMAND

→ Resuming engine from cycle 0
[engine] building kernel...
[engine] reusing externally-built kernel + hub (172-node actuator surface)
[boot] STATE.theta_hub mirrored from constructor_or_internal_build
[boot] hub config 'engine' registered — kernel constants ab first-class: 'set node engine/<attr> <value>'
[engine] kernel config (24 numeric attrs): AFR=14.7 · EVC=370.0 · EVO=130.0 · IVC=560.0 · IVO=340.0 · LHV=44000000.0 ...
[engine] (full values: 'system inventory' → KERNEL CONFIGURATION)
[engine] warming up (25 cycles)...
[engine] live — 5 cycles in buffer
```

EXPLANATION

What happened: Engine resumed from cycle 0. The Theta combustion kernel + hub are reused (172-node actuator surface), *theta_hub* is mirrored, and the 'engine' hub config is registered so 24 kernel constants become first-class settable nodes. Warm-up of 25 cycles, then live with 5 cycles buffered.

What this means: The physics simulator is the live data source. Its constants — AFR, valve timings (EVC/EVO/IVC/IVO), LHV — are now addressable, which is what makes later interventions possible. A '172-node actuator surface' is the controllable parameter space.

Evidence produced: Kernel config snapshot (AFR=14.7, valve angles, LHV=44 MJ/kg), 172-node actuator surface, buffer depth (5 cycles).

Why this matters: Without addressable actuators you cannot perturb-and-measure later. This step establishes the controllable substrate the whole closed loop depends on.

Insights: The engine starts nearly empty (5 cycles) — far too little to analyze yet. Data must accumulate before causal discovery is meaningful.

RAW LOG BLOCK — #2

```
[observer] ca50 std trend: 5.4→7.0° (+0.79°/interval, 3 consecutive increases)
[ROUTER] COMMAND
Active targets: ['ca50'] (focus = ca50)

[COMMAND] → Manual DIFC triggered (200 cycles in buffer)
Analysis running in background...
```

EXPLANATION

What happened: The read-only observer flagged ca50 standard-deviation rising **5.4→7.0°** (+0.79°/interval, 3 consecutive increases). Focus is set to ca50 and the user manually triggers DIFC over 200 buffered cycles; analysis runs in the background.

What this means: A drift detector noticed combustion-timing scatter growing — a destabilization signal — and a causal-discovery run was launched to find the cause. ca50 is the crank angle at which 50% of the charge has burned; rising std means combustion phasing is becoming erratic.

Evidence produced: ca50 std trend with slope (+0.79°/interval) and a 3-interval monotonic-increase flag.

Why this matters: This is the *observe* → *investigate* trigger of the closed loop in action — the system surfaces an anomaly and acts on it without being told what is wrong.

Insights: The symptom under investigation is variance in burn phasing, not a single bad value — a stability problem, which is harder and more interesting than a static offset.

RAW LOG BLOCK — #3

```
[difc-hb] 5% Initializing... (t+0.0s)
[difc-hb] 12% Data ready: 200 cycles × 41 vars (t+0.7s)
[difc-hb] 20% EXPERIMENT mode – causal discovery (200 cycles)... (t+0.9s)
[observer] DIFC[ca50] slow run (>25s) – abort NAHI, +35s grace (soft timeout)
[difc-hb] 55% Physics validation... (t+33.5s)
[difc-hb]   · pwe-prerefine start (t+33.5s)
[difc-hb]   · pwe-prerefine done (t+66.0s)
[difc-hb]   · filter-loop start (t+66.0s)
[observer] ca50: scope = 9 vars (7 DAG-ancestors + 2 data-prescreened) – 36 vars is run me discovery me nahi the (har run top-correlated rotate hote)
[observer] ca50: raw=36 valid=31 accepted=22 unknown=2 rejected=12 (stored=24: 22 confirmed + 2 provisional)
[observer] ca50: rejection reasons → 12× forbidden_direction
[difc-hb]   · filter-loop done (t+66.0s)
[difc-hb] 65% Sensitivity calibration... (t+66.0s)
[difc-hb] 72% SBL learning update... (t+98.1s)
[difc-hb] 80% ACIS reasoning... (t+98.2s)
[difc-hb] 87% Querying per-target decisions... (t+98.2s)
[difc-hb] 93% Multi-objective tradeoff... (t+98.2s)
[difc-hb] 97% Building causal graph... (t+98.2s)
[difc-hb]   · paths-extract start (t+98.2s)
[difc-hb]   · paths-extract done (t+98.2s)
[difc-hb]   · broadcast start (t+98.2s)
[difc-hb]   · bcast: payload built (t+98.2s)
[difc-hb]   · cdifc: entry (t+98.2s)
[difc-hb]   · cdifc: edge_memory done (t+98.2s)
[difc-hb]   · cdifc: freeze+carry done (t+98.3s)
[difc-hb]   · cdifc: rel-verify deferred (background) (t+98.3s)
[difc-hb]   · cdifc: census done (t+98.3s)
[difc-hb]   · cdifc: knowledge recomputed (t+98.3s)
[difc-hb]   · bcast: complete_difc done (t+98.3s)
[difc-hb]   · bcast: difc_done emitted (t+98.3s)
[difc-hb]   · broadcast done (t+98.3s)
[difc-hb] 100% Done – 24 edges (t+98.3s)

[observer] DIFC complete – causal structure updated
[observer] Economy: 5.5127 knowledge-value · discovery 20.031 CPU-sec · maintenance 97.19 CPU-sec
[observer] epistemic: U_total=0.38 · latent=0.50 (max; mean 0.36) (type 'ceh' for breakdown)
[ROUTER] CHAT

EBIS COGNITION MAP
Branch: evidence verification
Root: ca50 investigation
graph source: COMMITTED
reliability: VERIFYING (coverage 0/24 hops, background verify chal raha)
background verifiers: 1 active
```

EXPLANATION

What happened: First DIFC pass on 200×41 data in EXPERIMENT mode. It crossed the soft-timeout (>25s) and was granted +35s grace rather than aborting. PWE physics pre-refine + a filter loop ran; scope narrowed to **9 vars** (7 DAG-ancestors + 2 data-prescreened), 36 vars rotated out. Census: raw 36 → valid 31 → accepted 22, 2 unknown, **12 rejected (all forbidden_direction)**, stored 24 (22 confirmed + 2 provisional). SBL learning, ACIS reasoning, multi-objective tradeoff, graph build and broadcast follow; reliability-verify is deferred to background. Done in 98.3s → **24 edges**. Economy: knowledge-value 5.51, discovery 20.0 CPU-s, maintenance 97.2 CPU-s. Epistemics: U_total 0.38, latent 0.50 max / 0.36 mean.

What this means: This is a full causal-discovery run. *forbidden_direction* = candidate edges whose orientation violates the physics prior (e.g. effect→cause) are dropped by PWE/ACIS gating. *Scope rotation* means only a top-correlated subset enters discovery each run, so a variable's absence is not evidence of no relationship. *Provisional* edges are accepted but not yet evidence-backed.

Evidence produced: Full census (raw→valid→accepted/unknown/rejected→stored), rejection reason tally, edge count (24), a timestamped stage trace, economy triple, and the U_total / latent epistemic readout.

Why this matters: The census + rejection log is the audit trail proving edges were *filtered*, not invented. The soft-timeout shows the run chose completeness over speed; deferring reliability-verify shows the graph is committed before it is fully trusted.

Insights: 12 of 36 rejections being forbidden-direction is a large share — the physics prior is actively pruning. latent 0.50 says meaningful hidden-factor risk remains, so these edges are structural hypotheses, not confirmed physics.

RAW LOG BLOCK — #4

```
ca50
├─ ✓ ca10 +1.053 [conf 0.92 · unc 0.01] lag same-cyc trust:UNRELIABLE [~latent 0.47]
├─ ✓ mean flame speed -0.595 [conf 0.69 · unc 0.01] lag same-cyc trust:UNRELIABLE [~latent 0.47]
├─ ✓ volumetric efficiency -0.580 [conf 0.65 · unc 0.01] lag same-cyc trust:UNRELIABLE [~latent 0.50]
├─ +2 more drivers — type expand
└─
causes:
├─ ✓ ca10 +0.908 [conf 0.92 · unc 0.01] lag same-cyc [~latent 0.46]
└─
△ sibling-metric edge: ca10/ca50/ca90 ek hi burn-curve ke percentiles hain — co-output correlation, physical driver nahi
```

EXPLANATION

What happened: The committed ca50 cognition map. Reliability shows **VERIFYING (0/24 hops)** with 1 background verifier active. Drivers into ca50: ca10 (+1.053, conf 0.92), mean flame speed (-0.595, conf 0.69), volumetric efficiency (-0.580, conf 0.65), +2 more. ca50 causes ca10 (+0.908). A warning flags the ca10/ca50/ca90 edge as a sibling-metric (co-output) relationship.

What this means: Each edge carries a weight (standardized effect), *conf* (confidence), *unc* ($1-r^2$ unexplained variance), a lag tag (same-cycle), a trust tag, and a *latent* hidden-factor risk. **trust:UNRELIABLE** means no experiments back the edge yet. The sibling-metric warning is the system protecting itself: ca10/ca50/ca90 are three percentiles of the same burn curve, so their correlation is co-output, not a physical driver.

Evidence produced: Per-edge conf/unc/latent values, the verifying-coverage counter, and an explicit co-output caveat on the strongest edge.

Why this matters: Honest causal reading: the graph distinguishes a real driver from a sibling artifact and labels edges UNRELIABLE until experiments exist — it refuses to overclaim correlation as causation.

Insights: The biggest-weight edge (ca10) is precisely the one flagged as non-physical. The genuinely informative drivers are mean flame speed and volumetric efficiency, both physically plausible influences on burn phasing.

RAW LOG BLOCK — #5

```
[4 of 6 DIRECT edges shown (type expand) · CEH E = full scope graph, not just focus-edges]
+16 edges among other scope vars (not ca50-direct) — type 'show details' for full graph
legend: [conf · unc U] U=edge uncertainty ( $1-r^2$ , unexplained variance) or 4-component when present (~U = 1-conf proxy) latent L = hidden-factor risk
```

EXPLANATION

Cognition-map footer / legend (continuation of the ca50 map above). Only 4 of 6 direct edges are shown; +16 edges exist among other scope variables (not ca50-direct). The legend defines the display grammar used throughout: *conf* = confidence, *U* = edge uncertainty ($1-r^2$, or a 4-component form when present), and *latent L* = hidden-factor risk. CEH 'E' counts the full scope graph, not just the focus edges. This is presentation metadata, not a new computation.

RAW LOG BLOCK — #6

```
[ROUTER] CHAT

evidence verification
volumetric_efficiency causes:
├─ mean_flame_speed: weight=+0.9426 conf=0.88 [?]
├─ prev_residual_mass_kg: weight=-0.8648 conf=0.75 [?]
├─ ca50: weight=-0.5799 conf=0.78 [?]
├─ ca10: weight=-0.3682 conf=0.61 [?]
├─ burn_duration: weight=-0.3494 conf=0.70 [?]
└─
volumetric_efficiency is driven by:
├─ residual_fraction: weight=-0.8336 conf=0.92
├─ turbulence_intensity: weight=-0.7811 conf=0.85
├─ knock_elevated (29.72) — check causal drivers
├─ trust(ca50)=0.00 · weakest: evidence=0.00 · recover: run more experiments
```

EXPLANATION

What happened: A chat read of volumetric_efficiency relationships from the *existing* committed graph (before its own DIFC run). It causes mean_flame_speed (+0.943), prev_residual_mass (-0.865), ca50 (-0.580), ca10 (-0.368), burn_duration (-0.349); it is driven by residual_fraction (-0.834) and turbulence_intensity (-0.781). Side notes: knock elevated (29.72) and trust(ca50)=0.00 with a recovery hint.

What this means: The chat layer reads governed evidence only — it reports what the current graph already holds without mutating it. The [?] tags mark the 'causes' edges as not yet reliability-verified. The trust line restates that ca50 has zero experimental backing.

Evidence produced: A directional driver/effect list with weights and confidences, plus live knock and trust-deficit advisories.

Why this matters: Demonstrates the read/write separation: explanation is served from committed truth, and the system keeps surfacing its own weak spots (trust=0) rather than hiding them.

Insights: volumetric_efficiency sits upstream of flame speed and downstream of residual fraction / turbulence — a coherent breathing-vs-combustion story worth a dedicated discovery run, which is what happens next.

RAW LOG BLOCK — #7

```
[observer] New engine cycles arrived – causal model now stale (re-run DIFC to refresh)
[ROUTER] COMMAND
Active targets: ['volumetric_efficiency'] (focus = volumetric_efficiency)

[COMMAND] → Manual DIFC triggered (200 cycles in buffer)
Analysis running in background...

[difc-hb] 5% Initializing... (t+0.0s)
```

EXPLANATION

New engine cycles arrived, so the observer marks the causal model **stale** and the user triggers a fresh DIFC run focused on *volumetric_efficiency* (200 cycles). The run begins (5% initializing). Staleness here means the world has moved past the graph; the trigger is the *update* step of the loop firing.

RAW LOG BLOCK — #8

```
[difc-hb] 12% Data ready: 200 cycles × 40 vars (t+0.6s)
[difc-hb] 20% EXPERIMENT mode – causal discovery (200 cycles)... (t+0.6s)
[observer] reliability verified: ca50 – coverage 24/24 hops (73.7s background)
[ROUTER] COMMAND

evidence verification
```

EXPLANATION

What happened: Concurrency in action: while the new volumetric run is at 12→20%, a *background* reliability check completes for the previous target — **ca50 coverage 24/24 hops (73.7s)**. The user then stops the engine; the buffer freezes. Stale/knock advisories print and the volumetric cognition-map header is emitted.

What this means: Reliability verification re-walks every hop of a committed graph to confirm it reproduces; 24/24 means full coverage with no failed hops. This runs asynchronously, so discovery and verification overlap. 'edges stale (190 cycles old)' quantifies how far ca50's graph has drifted.

Evidence produced: A completed background-verify coverage figure (24/24) with wall-clock cost (73.7s), plus a frozen-buffer state marker.

Why this matters: Verification is decoupled from discovery — the system can keep working while independently auditing past results, and it timestamps drift so nothing silently goes stale.

Insights: ca50's graph fully verified *after* commit, confirming the earlier deferral was safe; meanwhile the engine is paused, so volumetric analysis runs on a fixed snapshot.

RAW LOG BLOCK — #9

```
→ Engine stopped
user request
```

EXPLANATION

Engine-stopped sub-banner (continuation). Records the halt as a user request inside the volumetric map context — provenance bookkeeping, no new analysis.

RAW LOG BLOCK — #10

```
Engine halted. Buffer frozen at last state.
~ knock elevated (28.50) — check causal drivers
~ ca50: edges stale (190 cycles old) — re-run DIFC
```

```
EBIS COGNITION MAP
Branch: evidence verification
Root: volumetric investigation
graph source: COMMITTED
```

EXPLANATION

Halt confirmation + volumetric map header (continuation). Buffer frozen at last state; knock (28.50) and ca50-staleness (190 cycles) advisories repeat. Sets up the volumetric cognition map that follows.

RAW LOG BLOCK — #11

```
volumetric
├─ ✓ residual fraction -0.834 [conf 0.82 · unc 0.09] lag same-cyc [latent 0.18]
├─ ✓ turbulence intensity -0.781 [conf 0.55 · unc 0.09] lag same-cyc [latent 0.33]
├─
├─ causes:
├─ ✓ prev residual mass kg -0.865 [conf 0.76 · unc 0.03] lag same-cyc [latent 0.17]
├─ ✓ mean flame speed +0.943 [conf 0.67 · unc 0.09] lag same-cyc [latent 0.42]
├─ ✓ burn duration -0.349 [conf 0.66 · unc 0.26] lag same-cyc [latent 0.17]
├─ +2 more effects — type expand
├─
├─ △ Tradeoff detected:
├─ ↑ mean flame speed +0.943 (improves)
├─ ↓ burn duration -0.349 (worsens)
├─ ↓ ca10 -0.368 (worsens)
```

EXPLANATION

What happened: The volumetric_efficiency cognition map. Drivers: residual fraction (-0.834, conf 0.82, latent 0.18), turbulence intensity (-0.781, conf 0.55, latent 0.33). Effects: prev residual mass (-0.865), mean flame speed (+0.943, latent ~0.42), burn duration (-0.349), +2 more. A **tradeoff is detected**: raising mean flame speed (+0.943, improves) worsens burn duration (-0.349) and ca10 (-0.368).

What this means: Lower latent values (0.18, 0.33) here mean these edges have less hidden-confounder risk than the ca50 sibling edges — a structurally cleaner subgraph. The tradeoff node is multi-objective reasoning: one lever helps one target while hurting others, so no free optimization exists.

Evidence produced: Driver/effect weights with conf/unc/latent, and an explicit tradeoff triple naming the competing objectives.

Why this matters: Surfacing a tradeoff is the difference between a correlation map and a decision-support tool — it tells an engineer that pushing flame speed has a cost elsewhere.

Insights: Residual fraction and turbulence are the controllable upstream knobs for volumetric efficiency; flame speed is the coupled variable that links breathing to combustion quality.

RAW LOG BLOCK — #12

```
Session: 1 exchanges
· volumetric efficiency
[5 of 7 DIRECT edges shown (type expand) · CEH E = full scope graph, not just focus-edges]
+15 edges among other scope vars (not volumetric-direct) — type `show details` for full graph
legend: [conf · unc U] U=edge uncertainty (1-r2, unexplained variance) or 4-component when present (~U = 1-conf proxy) latent L = hidden-factor risk
```

EXPLANATION

Volumetric map footer / session line (continuation). 5 of 7 direct edges shown; +15 edges among other scope vars; session has 1 exchange. Same legend grammar as before — display metadata only.

RAW LOG BLOCK — #13

```
[difc-hb] 55% Physics validation... (t+25.4s)
[difc-hb] · pwe-prerefine start (t+25.4s)
[difc-hb] · pwe-prerefine done (t+35.4s)
[difc-hb] · filter-loop start (t+35.4s)
[observer] volumetric efficiency: scope = 9 vars (3 DAG-ancestors + 6 data-prescreened) — 36 vars is run me discovery me nahi the (har run top-correlated rotate hote)
[observer] volumetric_efficiency: raw=28 valid=21 accepted=20 unknown=2 rejected=6 (stored=22: 20 confirmed + 2 provisional)
[observer] volumetric_efficiency: rejection reasons → 5x insufficient_evidence · 1x forbidden_direction
[difc-hb] · filter-loop done (t+35.4s)
[difc-hb] 65% Sensitivity calibration... (t+35.4s)
[difc-hb] 72% SBL learning update... (t+43.6s)
[difc-hb] 80% ACIS reasoning... (t+43.6s)
```

```
[dffc-hb] 87% Querying per-target decisions... (t+43.6s)
[dffc-hb] 93% Multi-objective tradeoff... (t+43.6s)
[dffc-hb] 97% Building causal graph... (t+43.6s)
[dffc-hb]   · paths-extract start (t+43.6s)
[dffc-hb]   · paths-extract done (t+43.6s)
[dffc-hb]   · broadcast start (t+43.6s)
[dffc-hb]   · bcast: payload built (t+43.6s)
[dffc-hb]   · cdiff: entry (t+43.6s)
[dffc-hb]   · cdiff: edge_memory done (t+43.6s)
[dffc-hb]   · cdiff: freeze+carry done (t+43.6s)
[dffc-hb]   · cdiff: rel-verify deferred (background) (t+43.6s)
[dffc-hb]   · cdiff: census done (t+43.6s)
[dffc-hb]   · cdiff: knowledge recomputed (t+43.6s)
[dffc-hb]   · bcast: complete_dffc done (t+43.6s)
[dffc-hb]   · bcast: dffc_done emitted (t+43.6s)
[dffc-hb]   · broadcast done (t+43.6s)
[dffc-hb] 100% Done — 22 edges (t+43.6s)

[observer] DIFC complete — causal structure updated
[observer] Economy: 11.369 knowledge-value · discovery 30.178 CPU-sec · maintenance 150.065 CPU-sec
[observer] epistemic: U_total=0.45 · latent=1.00 (max; mean 0.64) (type 'ceh' for breakdown)
[observer] reliability verified: volumetric_efficiency — coverage 22/22 hops (18.3s background)
[ROUTER] CHAT

— EBIS RUNTIME STATE —
organism:      blocked_by_code NO · blocked_by_evidence YES · 2 open · top: graph drift validation (type 'status')
engine:        offline
graph:         VALID · age 0 cyc
active targets: ['volumetric_efficiency']
buffer vars:   44 numeric · 4 low-variance
edges:         21 valid · 0 unknown · 7 rejected
census [ca50]: raw 36 → valid 31 · accepted 22 · unknown 2 · rejected 12 → stored 24 (22 confirmed + 2 provisional)
census [volumetric_efficiency]: raw 28 → valid 21 · accepted 20 · unknown 2 · rejected 6 → stored 22 (20 confirmed + 2 provisional)
experiments:   0 recorded
epistemics [volumetric_efficiency]: trust(T)=0.00 · latent(L)=1.00 · uncertainty(U_total)=0.45
economy:       0.0631 know-value/CPU-sec · 180.246 CPU-sec · 11.369 know-value

— WHAT YOU CAN DO NOW —
> engine chalao (start the engine)
> signal inventory (all buffer vars + health)
> system inventory (kya exist karta · kya connected — connection audit)
> show valid|unknown|rejected (edge explorer)
> graph evolution (how edges changed across runs)
> timeline (investigation history)
> freshness (engine-vs-graph age)
> trust evidence (trust store breakdown)
> efficiency (knowledge per compute · DCU)
> glossary (what knowledge/efficiency/trust/signal mean)
> ab kya karu (narrator recommends next step)

— HELP BRANCHES (type to open) —
help commands COMMANDS (engine + analysis)
help system SYSTEM & RUNTIME (status · health · integrity)
help explore EXPLORE (read the causal graph)
help experiments EXPERIMENTS (perturb + measure)
help control CONTROL (act on the engine)
help trust TRUST & ECONOMY
help narration NARRATION (ask the organism)
help glossary GLOSSARY
```

EXPLANATION

What happened: The volumetric DIFC run completes (55→100%): PWE pre-refine, scope = 9 vars (3 DAG-ancestors + 6 data-prescreened), census raw 28 → valid 21 → accepted 20, 2 unknown, 6 rejected (5 insufficient_evidence + 1 forbidden_direction), stored 22 (20+2). Done in 43.6s → **22 edges**. Economy 11.37 know-value, U_total 0.45, **latent 1.00 max** / 0.64 mean; reliability verified 22/22 (18.3s). Then a full RUNTIME STATE panel prints: organism blocked_by_evidence YES (2 open), engine offline, graph VALID age 0, 44 numeric vars, 21 valid / 7 rejected edges, 0 experiments, trust 0.00, economy 0.063 know-value/CPU-s.

What this means: *insufficient_evidence* rejections (new this run) mean some candidate edges lacked enough data, distinct from physics-illegal ones. **latent 1.00 max** is the worst hidden-factor risk seen so far — one edge is maximally suspect. *blocked_by_evidence* (not *by_code*) is the key organism distinction: the system is not broken, it is waiting for experiments to raise trust.

Evidence produced: Second full census, a new rejection category, the economy/epistemic triple, and a complete runtime-state snapshot including the trust deficit (0 experiments).

Why this matters: The runtime-state panel is the single-pane forensic summary — it ties graph validity, census, trust and economy together and names *why* progress is blocked (evidence, not code).

Insights: Despite 22 verified edges, trust is still 0.00 because zero experiments have run — the system correctly refuses to convert structure into confidence without interventional data. latent 1.00 flags exactly where to look.

RAW LOG BLOCK — #14

```
[ROUTER] CHAT

— HELP · SYSTEM & RUNTIME (status · health · integrity) —
```



```

status / health / diagnostics      full runtime state: organism, engine, graph, census, economy
organism status / organism health  blocked-by-code vs blocked-by-evidence + open threads
event log / runtime governance     background threads, recent events, errors
boot check / integrity             runtime integrity: env, sys.path, orch lifecycle, signature drift
show errors / log                 recent errors + DIFC trace log for this session
why difc failed                   why the last DIFC run produced nothing (lifecycle + reason + source)
snapshots                        snapshot archive + last DIFC result lineage
repair report                     data-repair audit for the current session
staleness / difc age              engine-vs-graph age (how stale the causal model is)

(back to index: 'help')

```

EXPLANATION

HELP · SYSTEM & RUNTIME menu. A static index of diagnostic commands (status, integrity/boot check, event log, why-difc-failed, snapshots, repair report, staleness). No computation — it documents the introspection surface the operator can call.

RAW LOG BLOCK — #15

```

[ROUTER] CHAT
No data repairs in current session — all columns clean.

```

EXPLANATION

Repair report: 'No data repairs in current session — all columns clean.' The data-repair auditor ran and found nothing to fix — every buffer column passed validation. A clean negative result, which is itself evidence of input integrity.

RAW LOG BLOCK — #16

```

[ROUTER] CHAT
No data repairs in current session — all columns clean.

```

EXPLANATION

Repair report (repeated invocation). Same clean result — no repairs needed. Confirms the previous check was not a fluke; the buffer remains uncorrupted.

RAW LOG BLOCK — #17

```

[ROUTER] CHAT

- DIFC FRESHNESS -
engine now (cycle):    585
DIFC built at (cycle): 585
graph age (cycles):    0  ✓ fresh
lifecycle:             VALID (running=False)
  volumetric_efficiency built@585 age=0
    ca50                built@395 age=190

→ Narration/ASCII/trust/predictions all read THIS graph.
→ If age is large, they describe a world the engine has left.

```

EXPLANATION

What happened: DIFC FRESHNESS panel: engine now at cycle 585, volumetric graph built@585 (age 0, fresh), ca50 built@395 (age 190). Lifecycle VALID, running=False. Notes that narration / ASCII / trust / predictions all read THIS graph, and that large age means they describe a world the engine has left.

What this means: Freshness is per-target: volumetric is current, ca50 is 190 cycles behind. Everything downstream (predictions, narration) consumes the committed graph, so its age directly bounds how trustworthy those outputs are.

Evidence produced: Per-target build-cycle and age values, plus an explicit dependency note that all reads inherit graph age.

Why this matters: Staleness is a first-class, measured quantity here, not an afterthought — the system tells you exactly how out-of-date each causal model is before you rely on it.

Insights: Any ca50 prediction made now is reading a 190-cycle-old structure; the panel is effectively warning that ca50 conclusions need a refresh before trust.

RAW LOG BLOCK — #18

[ROUTER] CHAT

evidence verification

EXPLANATION

Prediction context header (continuation). Opens a volumetric-efficiency prediction under the evidence-verification branch — framing for the forecast block that follows.

RAW LOG BLOCK — #19

```
→ Prediction: volumetric efficiency
horizon=15 cycles · forecast value = time-series (statistical); ORCH/DIFC sections below are context, not in the number
```

EXPLANATION

Prediction banner (continuation). Declares horizon = 15 cycles and states plainly that the forecast value is a *statistical time-series* projection; the ORCH/DIFC sections shown are context, not inputs to the number. This separation is the honesty guard — structural drivers explain but do not silently inflate the forecast.

RAW LOG BLOCK — #20

```
current value: +0.847
recent trend:  stable (-0.0004/cycle)

Projection (reversion × 15 cycles):
volumetric efficiency: +0.847 → +0.896 Δ=+0.049 [moderate confidence]
basis: mean-reversion (oscillating; mean +0.90)

DIFC structural drivers of volumetric efficiency:
indicated → volumetric efficiency effect=+1.104/unit
(9 self/sibling/duplicate edge(s) hidden – co-output, not physical driver)
basis: DIFC causal graph, single-regime estimate [explanatory – NOT in forecast value above]

Predictions improve with each experiment (ORCH learning).
Trend projection assumes no parameter changes.
~ knock elevated (28.50) – check causal drivers
~ ca50: edges stale (190 cycles old) – re-run DIFC
```

EXPLANATION

What happened: Volumetric-efficiency forecast: current +0.847, trend stable (-0.0004/cycle). Projection over 15 cycles → **+0.896 (Δ=+0.049, moderate confidence)**, basis mean-reversion toward +0.90. DIFC structural drivers are shown separately (indicated → volumetric, +1.104/unit), with 9 self/sibling/duplicate edges hidden as co-output.

What this means: Two independent things are reported side by side: a statistical mean-reversion forecast (the actual number) and a structural-driver explanation (context only). The system reverts an oscillating signal toward its mean rather than extrapolating the tiny slope.

Evidence produced: A point forecast with delta and confidence band, the reversion basis (mean +0.90), and a separated, co-output-filtered driver list.

Why this matters: Keeping the forecast value and the causal explanation in separate compartments prevents structure from contaminating the prediction — a disciplined separation many ML pipelines blur.

Insights: The prediction is modest and mean-reverting; with 9 co-output edges suppressed, only 'indicated work' survives as a non-sibling structural driver, so the explanatory signal is thin.

RAW LOG BLOCK — #21

```
[ROUTER] COMMAND
Active targets: ['ca50'] (focus = ca50)

[COMMAND] → Manual DIFC triggered (200 cycles in buffer)
Analysis running in background...

[difc-hb] 5% Initializing... (t+0.0s)
```

EXPLANATION

DIFC trigger — focus ca50 (200 cycles). The operator re-runs discovery on ca50, presumably to refresh its 190-cycle-old graph. Run begins at 5%.

RAW LOG BLOCK — #22

```
[difc-hb] 12% Data ready: 200 cycles × 40 vars (t+0.1s)
[difc-hb] 20% EXPERIMENT mode – causal discovery (200 cycles)... (t+0.1s)
[difc-hb] 55% Physics validation... (t+8.4s)
[difc-hb] · pwe-prerefine start (t+8.4s)
[difc-hb] · pwe-prerefine done (t+14.5s)
[difc-hb] · filter-loop start (t+14.5s)
[observer] ca50: scope = 9 vars (7 DAG-ancestors + 2 data-prescreened) – 36 vars is run me discovery me nahi the (har run top-correlated
rotate hote)
[observer] ca50: raw=27 valid=15 accepted=14 unknown=4 rejected=9 (stored=18: 14 confirmed + 4 provisional)
[observer] ca50: rejection reasons → 9× forbidden_direction
[difc-hb] · filter-loop done (t+14.5s)
[difc-hb] 65% Sensitivity calibration... (t+14.5s)
[difc-hb] 72% SBL learning update... (t+17.9s)
[difc-hb] 80% ACIS reasoning... (t+17.9s)
[difc-hb] 87% Querying per-target decisions... (t+17.9s)
[difc-hb] 93% Multi-objective tradeoff... (t+17.9s)
[difc-hb] 97% Building causal graph... (t+17.9s)
[difc-hb] · paths-extract start (t+17.9s)
[difc-hb] · paths-extract done (t+17.9s)
[difc-hb] · broadcast start (t+17.9s)
[difc-hb] · bcast: payload built (t+17.9s)
[difc-hb] · cdifc: entry (t+17.9s)
[difc-hb] · cdifc: edge_memory done (t+17.9s)
[difc-hb] · cdifc: freeze+carry done (t+18.0s)
[difc-hb] · cdifc: rel-verify deferred (background) (t+18.0s)
[difc-hb] · cdifc: census done (t+18.0s)
[difc-hb] · cdifc: knowledge recomputed (t+18.0s)
[difc-hb] · bcast: complete difc done (t+18.0s)
[difc-hb] · bcast: difc_done emitted (t+18.0s)
[difc-hb] · broadcast done (t+18.0s)
[difc-hb] 100% Done – 18 edges (t+18.0s)

[observer] DIFC complete – causal structure updated
[observer] Economy: 14.6025 knowledge-value · discovery 38.364 CPU-sec · maintenance 150.065 CPU-sec
[observer] epistemic: U_total=0.37 · latent=0.71 (max; mean 0.48) (type 'ceh' for breakdown)
[ROUTER] CHAT

– CEH SIGNALS [ca50] (each raw, separate – no composite) –
E edges (valid) : 15 [13 physical · 2 sibling/co-output (ca10/ca50/ca90 – not physical)]
T trust (maturity) : 0.00 [freshness 1.00 · evidence 0.00]
Ev evidence (experiments) : 0
K knowledge-value : 14.6025
L latent score (max) : 0.71 [mean 0.48 · n=18 · carrier ('ca50', 'ca90')]
note: L MAX-aggregate hai – edges badhne se sirf badh sakta hai (knowledge ke saath L contradiction nahi)
Ue edge unc (1-r², mean) : 0.056 [unexplained variance, [0,1]]
oe coef stability (mean) : 0.1183 [bootstrap coef-std · n=18 · single-window estimate, NOT in U_total]
Ug uncertainty_global : 0.056 [latent-reconstruction, when present]
P prediction unc : n/a [no error_memory yet]
Uo observability unc : 0.225
Us structure unc (rej/tot) : 0.333
Ur regime unc (1-fresh) : 0.000

U_total (aggregate) : 0.371 [components present: U_edge,U_obs,U_structure,U_trust,U_regime +latent]

note: rows are RAW separate signals (oe = edge-estimate stability,
NOT reality uncertainty). U_total = weighted aggregate of present components.
```

EXPLANATION

What happened: A fast ca50 re-run: done in just 18.0s → **18 edges** (vs 98.3s earlier). Census raw 27 → valid 15 → accepted 14, 4 unknown, 9 rejected (forbidden_direction), stored 18 (14+4). Economy 14.60 know-value, U_total 0.37, latent 0.71 max / 0.48 mean. Then the **CEH SIGNALS [ca50]** breakdown prints every uncertainty component raw and separate: E=15 edges (13 physical, 2 sibling), T=0.00 trust (freshness 1.00, evidence 0.00), Ev=0 experiments, K=14.60, L=0.71 (max-aggregate), Ue=0.056 edge unc, σ_e =0.118 coef stability, Uo=0.225 observability, Us=0.333 structure (rej/total), Ur=0.000 regime, P=n/a; U_total=0.371.

What this means: The same target now resolves far faster because warm caches and prior edge-memory cut discovery cost. The CEH panel decomposes uncertainty into independent sources rather than one opaque number: edge fit (Ue), coefficient stability (σ_e), observability (Uo), structural rejection rate (Us), regime freshness (Ur), trust, and latent. The note that L is a MAX-aggregate explains why adding edges can only raise L — it is not a contradiction with rising knowledge.

Evidence produced: A second ca50 census, full per-component uncertainty vector, and the explicit statement that σ_e is estimate-stability, NOT reality uncertainty.

Why this matters: This is the heart of the epistemic honesty design: U_total is shown as a weighted sum of *named, separately auditable* parts, so a reviewer can see exactly which kind of uncertainty dominates (here structure, Us=0.333).

Insights: Structure uncertainty (1/3 of candidates rejected) and trust (0, no experiments) dominate ca50's uncertainty — not noisy fits. The cure is experiments and more regimes, not more correlation.

RAW LOG BLOCK — #23

```
[observer] reliability verified: ca50 – coverage 18/18 hops (31.9s background)
[ROUTER] COMMAND
```

```

→ Resuming engine from cycle 585
[engine] building kernel...
[engine] reusing externally-built kernel + hub (172-node actuator surface)
[engine] kernel config (24 numeric attrs): AFR=14.7 · EVC=370.0 · EVO=130.0 · IVC=560.0 · IVO=340.0 · LHV=44000000.0 ...
[engine] (full values: 'system inventory' → KERNEL CONFIGURATION)
[engine] warming up (25 cycles)...
[engine] live — 585 cycles in buffer

```

EXPLANATION

What happened: Background reliability verification confirms **ca50 coverage 18/18 hops (31.9s)**. The engine is then resumed from cycle 585; kernel/hub reused, warm-up 25 cycles, live with 585 cycles buffered.

What this means: The freshly re-run ca50 graph reproduces fully under independent re-walking. Resuming from 585 (not 0) means the engine continues the same timeline, so new data extends the existing buffer rather than restarting it.

Evidence produced: A 18/18 verification coverage figure with cost, and a buffer-depth confirmation (585 cycles).

Why this matters: Re-verification after the fast re-run shows the speed-up did not cost reliability — the 18-edge graph is as fully covered as the slow one was.

Insights: ca50 is now both fresh and verified, and the engine has a deep buffer (585 cycles), setting up meaningful prediction-versus-actual checks.

RAW LOG BLOCK — #24

```

[verify] prediction volumetric: predicted=+0.9 actual=+1.0 error=0.1 ✓
[ROUTER] CHAT

△ variable: ca50 DIFC scope: 18 valid edges signal=0.34 difc=VALID

△ DIFC: 1 drivers 2 effects signal=0.34

drivers
ca50 signal=0.34 [MODERATE]

↳ caused by (evidence weight):
  residual  +0.468 DORMANT △ ~lat 0.43 unc 0.00 trust:UNRELIABLE seen=1x
  residual → ca50: no meaningful effect

→ ca50 propagates to:
  ca90  +1.002 WEAK
  ignition  +0.854 WEAK

Ca50 consistently increases ca90 (limited evidence, single-regime).

start investigating:
· ca50 hide karo — isolate its effect
· ascii — causal map
~ knock elevated (28.74) — check causal drivers

EBIS COGNITION MAP
Branch: drivers investigation
Trail: evidence verification → drivers investigation
Root: volumetric investigation
graph source: COMMITTED
reliability: VERIFIED (coverage 18/18 hops)

```

EXPLANATION

What happened: Prediction verification — volumetric: predicted +0.9, actual +1.0, error 0.1 ✓ (hit). Then a ca50 driver chat: signal 0.34 [MODERATE], difc VALID, 1 driver / 2 effects. Driver: residual (+0.468, DORMANT, ~lat 0.43, trust UNRELIABLE, seen 1x) — annotated 'no meaningful effect'. Effects: ca50 → ca90 (+1.002, WEAK), ca50 → ignition (+0.854, WEAK).

What this means: The earlier forecast is now checked against reality and *passed* within tolerance — the loop closes. In the driver view, DORMANT + 'seen 1x' + 'no meaningful effect' means the residual→ca50 edge exists structurally but has negligible, barely-observed influence. The bar glyphs encode relative evidence weight.

Evidence produced: A logged prediction-vs-actual pair with signed error and pass mark, plus an evidence-weighted driver readout with dormancy and trust flags.

Why this matters: Closed-loop validation is the system's core claim — here it is demonstrated end to end: a forecast was made, stored, and scored against the engine's later real value.

Insights: ca50's only inbound driver is effectively inert; ca50 acts mainly as an *output* propagating to ca90/ignition (both sibling/co-output, hence WEAK). Real upstream control of ca50 is still unidentified.

RAW LOG BLOCK — #25

```

ca50
├─ ← x residual fraction +0.468 [conf 0.40 · unc 0.00] lag same-cyc trust:UNRELIABLE [~latent 0.43]
└─
causes:

```

```

└─→ ✓ ca90 +1.002 [conf 0.93 · unc 0.00] lag same-cyc [~latent 0.71]
└─→ ✓ ignition delay index +0.854 [conf 0.89 · unc 0.02] lag same-cyc [~latent 0.68]
△ sibling-metric edge: ca10/ca50/ca90 ek hi burn-curve ke percentiles hain – co-output correlation, physical driver nahi

```

EXPLANATION

What happened: The ca50 driver cognition map. Reliability VERIFIED (18/18). Inbound: residual fraction X +0.468 (conf 0.40, trust UNRELIABLE, ~latent 0.43) — the X marks a rejected/non-confirmed driver. Outbound: ca50 → ca90 (+1.002, conf 0.93, ~latent 0.71), ca50 → ignition delay index (+0.854, conf 0.89). Sibling-metric warning repeats for ca10/ca50/ca90.

What this means: The X versus ✓ glyphs distinguish confirmed from non-confirmed edges in the same map. ca50's strong outbound edges land on its own burn-curve siblings (ca90) and on ignition delay, so high weight does not imply a usable causal lever.

Evidence produced: A mixed confirmed/rejected edge view with conf/latent, and the recurring co-output caveat.

Why this matters: The map is explicit about what it does *not* trust (the rejected residual edge) while showing the verified outbound structure — transparency about both halves.

Insights: Confirms the prior insight: ca50 is phasing output, not a controllable input. To stabilize ca50's variance you must reach further upstream (breathing, turbulence, residual dynamics), which the sibling edges cannot give you.

RAW LOG BLOCK — #26

```

Session: 3 exchanges
· volumetric efficiency
· engine stop
· predictions
[3 direct edges shown – all evidence-backed · CEH E = full scope graph]
+36 edges among other scope vars (not ca50-direct) – type `show details` for full graph
legend: [conf · unc U] U=edge uncertainty (1-r2, unexplained variance) or 4-component when present (~U = 1-conf proxy) latent L = hidden-
factor risk

```

EXPLANATION

ca50 driver-map footer (continuation). 3 direct edges shown (all evidence-backed); +36 edges among other scope vars; session 3 exchanges. Display metadata.

RAW LOG BLOCK — #27

```

[ROUTER] CHAT

– EBIS RUNTIME STATE –
organism:      blocked_by_code NO · blocked_by_evidence YES · 1 open · top: experiment-economy linkage (type 'status')
engine:        running cyc 705
graph:         VALID · age 120 cyc
active targets: ['ca50']
buffer vars:   44 numeric · 4 low-variance
edges:         15 valid · 1 unknown · 11 rejected
census [ca50]: raw 27 → valid 15 · accepted 14 · unknown 4 · rejected 9 → stored 18 (14 confirmed + 4 provisional)
census [volumetric_efficiency]: raw 28 → valid 21 · accepted 20 · unknown 2 · rejected 6 → stored 22 (20 confirmed + 2 provisional)
experiments:   0 recorded
epistemics [ca50]: trust(T)=0.00 · latent(L)=0.71 · uncertainty(U_total)=0.34
economy:       0.069 know-value/CPU-sec · 211.529 CPU-sec · 14.6025 know-value

– WHAT YOU CAN DO NOW –
> drivers dikhao (inspect current causes)
> full experiment karo (perturb + measure)
> signal inventory (all buffer vars + health)
> system inventory (kya exist karta · kya connected – connection audit)
> show valid|unknown|rejected (edge explorer)
> graph evolution (how edges changed across runs)
> timeline (investigation history)
> freshness (engine-vs-graph age)
> trust evidence (trust store breakdown)
> efficiency (knowledge per compute · DCU)
> glossary (what knowledge/efficiency/trust/signal mean)
> ab kya karu (narrator recommends next step)

– HELP BRANCHES (type to open) –
help commands  COMMANDS (engine + analysis)
help system    SYSTEM & RUNTIME (status · health · integrity)
help explore   EXPLORE (read the causal graph)
help experiments EXPERIMENTS (perturb + measure)
help control   CONTROL (act on the engine)
help trust     TRUST & ECONOMY
help narration NARRATION (ask the organism)
help glossary  GLOSSARY

```

EXPLANATION

What happened: RUNTIME STATE #2: organism blocked_by_evidence YES (1 open: experiment→economy linkage), engine running cyc 705, graph VALID **age 120 cyc**, active target ca50, 15 valid / 1 unknown / 11 rejected edges, 0 experiments, trust 0.00, latent 0.71, U_total 0.34, economy 0.069 know-value/CPU-s over 211.5 CPU-s.

What this means: A live snapshot mid-run: the engine has advanced to 705 while the ca50 graph is already 120 cycles old, and the single open organism thread is now about linking experiments to economic value. Trust remains 0 — still no interventions.

Evidence produced: A second full state vector: edge tally, census, epistemics, and a cumulative economy figure (211.5 CPU-s spent).

Why this matters: Repeated runtime-state panels make the audit trail self-checkpointing — each one is a comparable forensic snapshot you can diff against the last.

Insights: Even running, the graph is drifting (age 120) and trust is pinned at 0; the system's own open thread (experiment→economy) names the missing piece: interventions that would justify the compute spend.

RAW LOG BLOCK — #28

```
[observer] New engine cycles arrived – causal model now stale (re-run DIFC to refresh)
[ROUTER] CHAT

- BUFFER INVENTORY (44 numeric vars, target ca50) -
  ignition_delay_index      var=3.686      ✓6 edges
  burn_duration             var=0.9327     ✓3 edges
  spark_angle_deg           var=2.056      ✓3 edges
  turbulence_intensity       var=0.000253   ✓3 edges
  ca50                      var=9.361       ✓2 edges
  ca90                      var=9.312       ✓2 edges
  mean_flame_speed          var=0.04081    ✓2 edges
  volumetric_efficiency     var=0.001643   ✓1 edges
  air_fuel_ratio            var=0          Δlow-var ·no edges
  blowdown_intensity        var=0.03509    ·no edges
  breathing_efficiency       var=0.000586   ·no edges
  burn_acceleration         var=2.2e-05    ·no edges
  ca10                     var=5.257      ·no edges
  charge_density            var=0.002964   ·no edges
  combustion_stability       var=0.000123   ·no edges
  compression_heating       var=0.000445   ·no edges
  compression_heating_ratio var=0.000445   ·no edges
  dilution_penalty         var=0          ·no edges
  discharge_efficiency       var=0.000353   ·no edges
  engine_speed_rpm          var=0          Δlow-var ·no edges
  exhaust_backpressure_ratio var=0.003347   ·no edges
  exhaust_pressure_pa        var=3.493e+05  ·no edges
  exhaust_temperature_k      var=68.44      ·no edges
  exhaust_velocity          var=1.471      ·no edges
  flame_speed_peak          var=0.2094     ·no edges
  flow_stability_index       var=0.002021   ·no edges
  indicated_work_j           var=1098       ·no edges
  intake_temp_k             var=493.2      ·no edges
  intake_velocity           var=21.26      ·no edges
  knock_intensity           var=4.651      ·no edges
  manifold_pressure_pa       var=3.77e+07   ·no edges
  manifold_temp_k           var=395.3      ·no edges
  ncm_residual_mass_kg       var=0          Δlow-var ·no edges
  ncm_residual_pressure_pa   var=2.816e+05  ·no edges
  ncm_residual_temp_k        var=294.5      ·no edges
  peak_pressure_pa          var=2.793e+11  ·no edges
  prev_residual_mass_kg      var=0          ·no edges
  prev_residual_pressure_pa  var=1.511e+05  ·no edges
  prev_residual_temp_k       var=3.372e+04  ·no edges
  residual_fraction         var=0.00055    ·no edges
  rpm                      var=0          Δlow-var ·no edges
  scavenging_penalty         var=0.000134   ·no edges
  trapped_mass_kg           var=0          ·no edges
  volumetric_efficiency_real var=0.001643   ·no edges

Δ no-variation (variance-gated out of DIFC): ['air_fuel_ratio', 'engine_speed_rpm', 'ncm_residual_mass_kg', 'rpm']
· no valid edges yet: ['air_fuel_ratio', 'blowdown_intensity', 'breathing_efficiency', 'burn_acceleration', 'ca10', 'charge_density']

[observer] System: causal model stale – type 'rerun difc' to refresh
```

EXPLANATION

What happened: Model marked stale, then a full BUFFER INVENTORY of 44 numeric vars (target ca50). Edge-bearing vars: ignition_delay_index (6 edges), burn_duration / spark_angle / turbulence_intensity (3 each), ca50 / ca90 / mean_flame_speed (2 each), volumetric_efficiency (1). Four variance-gated out (var=0): air_fuel_ratio, engine_speed_rpm, ncm_residual_mass_kg, rpm. Many vars currently carry no edges.

What this means: DIFC only considers variables that actually move — zero-variance signals (constant AFR, rpm) are gated out because a constant cannot be a cause. Edge count per variable is a rough centrality measure: ignition_delay_index is the most connected node.

Evidence produced: A complete signal-health inventory with per-variable variance and edge counts, plus the explicit variance-gated and no-edge-yet lists.

Why this matters: This is the observability ledger — it shows exactly what the causal engine can and cannot 'see', and why (variance), which bounds every downstream claim.

Insights: ignition_delay_index is the structural hub of this regime. The held-constant actuators (AFR, rpm) explain why interventions are needed: nothing is varying them, so their causal role is currently invisible.

RAW LOG BLOCK — #29

```
[ROUTER] COMMAND
Active targets: ['intake_velocity'] (focus = intake_velocity)

[COMMAND] → Manual DIFC triggered (200 cycles in buffer)
Analysis running in background...

[difc-hb] 5% Initializing... (t+0.0s)
```

EXPLANATION

DIFC trigger — focus intake_velocity (200 cycles). The operator pivots to a breathing-side variable and starts a fresh discovery run (5% initializing).

RAW LOG BLOCK — #30

```
[difc-hb] 12% Data ready: 200 cycles × 40 vars (t+0.5s)
[difc-hb] 20% EXPERIMENT mode — causal discovery (200 cycles)... (t+0.5s)
[difc-hb] 55% Physics validation... (t+6.5s)
[difc-hb]   · pwe-prerefine start (t+6.5s)
[difc-hb]   · pwe-prerefine done (t+24.6s)
[difc-hb]   · filter-loop start (t+24.6s)
[observer] intake_velocity: scope = 9 vars (1 DAG-ancestors + 8 data-prescreened) — 36 vars is run me discovery me nahi the (har run top-
correlated rotate hote)
[observer] intake_velocity: raw=15 valid=15 accepted=15 unknown=0 rejected=0 (stored=15: 15 confirmed + 0 provisional)
[difc-hb]   · filter-loop done (t+24.6s)
[difc-hb] 65% Sensitivity calibration... (t+24.6s)
[difc-hb] 72% SBL learning update... (t+44.3s)
[difc-hb] 80% ACIS reasoning... (t+44.3s)
[difc-hb] 87% Querying per-target decisions... (t+44.3s)
[difc-hb] 93% Multi-objective tradeoff... (t+44.3s)
[difc-hb] 97% Building causal graph... (t+44.3s)
[difc-hb]   · paths-extract start (t+44.3s)
[difc-hb]   · paths-extract done (t+44.3s)
[difc-hb]   · broadcast start (t+44.3s)
[difc-hb]   · bcast: payload built (t+44.3s)
[difc-hb]   · cdifc: entry (t+44.3s)
[difc-hb]   · cdifc: edge_memory done (t+44.3s)
[difc-hb]   · cdifc: freeze+carry done (t+44.3s)
[difc-hb]   · cdifc: rel-verify deferred (background) (t+44.3s)
[difc-hb]   · cdifc: census done (t+44.3s)
[difc-hb]   · cdifc: knowledge recomputed (t+44.3s)
[difc-hb]   · bcast: complete difc done (t+44.3s)
[difc-hb]   · bcast: difc_done emitted (t+44.3s)
[difc-hb]   · broadcast done (t+44.3s)
[difc-hb] 100% Done — 15 edges (t+44.3s)

[observer] DIFC complete — causal structure updated
[observer] Economy: 19.3214 knowledge-value · discovery 40.883 CPU-sec · maintenance 246.382 CPU-sec
[observer] epistemic: U_total=0.25 · latent=0.66 (max; mean 0.34) (type 'ceh' for breakdown)
[ROUTER] CHAT

drivers
```

EXPLANATION

What happened: intake_velocity DIFC run: scope 9 vars (1 DAG-ancestor + 8 data-prescreened), census raw 15 → valid 15 → **accepted 15, 0 rejected**, stored 15 (all confirmed). Done in 44.3s → 15 edges. Economy 19.32 know-value, U_total **0.25** (lowest yet), latent 0.66 max / 0.34 mean. A prediction header for intake velocity is emitted.

What this means: A clean discovery: every candidate survived (0 rejections), unlike ca50's heavy forbidden-direction pruning. Lower U_total (0.25) means intake_velocity is a better-behaved, more identifiable target than the burn-phasing metrics.

Evidence produced: A zero-rejection census, the lowest U_total of the session, and the start of an intake-velocity forecast.

Why this matters: Contrast with ca50 is the insight: some variables are intrinsically easier to model. The census shows intake_velocity is one of them — fewer physics conflicts, cleaner structure.

Insights: The breathing side of the engine yields cleaner causal structure than the combustion-phasing side, suggesting intake_velocity may give more reliable predictions — about to be tested.

RAW LOG BLOCK — #31

```
→ Prediction: intake_velocity
horizon=15 cycles · forecast value = time-series (statistical); ORCH/DIFC sections below are context, not in the number
```

EXPLANATION

Intake-velocity prediction banner (continuation). Horizon 15 cycles; same honesty framing — forecast value is statistical, DIFC drivers are context only.

RAW LOG BLOCK — #32

```
current value: +109.203
recent trend:  stable (-0.0019/cycle)

Projection (reversion × 15 cycles):
  intake velocity: +109.203 → +104.445 Δ=-4.758 [high confidence]
  basis: mean-reversion (oscillating; mean +103.99)

DIFC structural drivers of intake velocity:
  manifold    → intake velocity    effect=-0.921/unit
  mean        → intake velocity    effect=-0.894/unit
  flame       → intake velocity    effect=-0.860/unit
  intake      → intake velocity    effect=-0.856/unit
(2 self/sibling/duplicate edge(s) hidden – co-output, not physical driver)
basis: DIFC causal graph, single-regime estimate [explanatory – NOT in forecast value above]

Predictions improve with each experiment (ORCH learning).
Trend projection assumes no parameter changes.
~ knock elevated (29.37) – check causal drivers
~ trust(ca50)=0.00 · weakest: evidence=0.00 · recover: run more experiments
~ ca50: edges stale (435 cycles old) – re-run DIFC
```

EXPLANATION

What happened: Intake-velocity forecast: current +109.203, trend stable (-0.0019/cycle). Projection → **+104.445 (Δ=-4.758, high confidence)**, basis mean-reversion toward +103.99. Structural drivers (context only): manifold (-0.921), mean flame speed (-0.894), flame (-0.860), intake (-0.856), with 2 co-output edges hidden.

What this means: A **high**-confidence forecast (vs 'moderate' for volumetric), consistent with this target's lower uncertainty. The reversion pulls the current value down toward the long-run mean of ~104.

Evidence produced: A point forecast with delta and a high-confidence band, plus a co-output-filtered driver list dominated by manifold/thermal terms.

Why this matters: The confidence label tracks the earlier U_{total} reading — the system's stated confidence is grounded in its measured uncertainty, not arbitrary.

Insights: Manifold and intake temperatures are the dominant structural influences on intake velocity — a physically sensible thermal-throttling story. The forecast predicts a ~4.8-unit drop.

RAW LOG BLOCK — #33

```
[ROUTER] CHAT

EBIS COGNITION MAP
Branch: drivers investigation
Trail: evidence verification → drivers investigation
Root:  intake investigation
graph source: COMMITTED
reliability: VERIFYING (coverage 0/15 hops, background verify chal raha)
background verifiers: 1 active
```

EXPLANATION

Intake cognition-map header (continuation). Reliability VERIFYING (0/15 hops), 1 background verifier active — the map below is committed but not yet re-walked.

RAW LOG BLOCK — #34

```
intake
├─ ✓ manifold temp k -0.921 [conf 0.90 · unc 0.06] lag same-cyc trust:UNRELIABLE [latent 0.06]
├─ ✓ intake temp k -0.856 [conf 0.86 · unc 0.06] lag same-cyc trust:UNRELIABLE [latent 0.06]
├─ ✓ mean flame speed -0.894 [conf 0.68 · unc 0.06] lag same-cyc trust:UNRELIABLE [latent 0.48]
├─ +1 more drivers – type expand
└─
causes:
├─ ✓ manifold temp k -0.926 [conf 0.90 · unc 0.01] lag same-cyc [latent 0.12]
├─ ✓ intake temp k -0.853 [conf 0.84 · unc 0.00] lag same-cyc [latent 0.17]
```

EXPLANATION

What happened: Intake_velocity cognition map. Drivers: manifold temp (-0.921, conf 0.90, **latent 0.06**), intake temp (-0.856, latent 0.06), mean flame speed (-0.894, ~latent 0.48), +1 more. Effects: manifold temp (-0.926), intake temp (-0.853), both very low latent.

What this means: **latent 0.06** is the cleanest seen in the session — these thermal driver edges carry almost no hidden-confounder risk, unlike the ca50 sibling edges (latent 0.47–0.71). Low unc (0.00–0.06) reinforces a tight fit.

Evidence produced: Driver/effect edges with notably low latent and unc, indicating a well-identified subgraph.

Why this matters: This is the system's most physically trustworthy map so far — low latent + high confidence + clean census — a model of what a 'good' EBIS result looks like.

Insights: Intake velocity is tightly coupled to manifold/intake temperature with little hidden confounding; this subgraph is the strongest candidate for reliable prediction and, eventually, intervention.

RAW LOG BLOCK — #35

```
Session: 5 exchanges
· predictions
· single inventory
· predict intake_velocity
[5 of 6 DIRECT edges shown (type expand) · CEH E = full scope graph, not just focus-edges]
+48 edges among other scope vars (not intake-direct) – type `show details` for full graph
legend: [conf · unc U] U=edge uncertainty (1-r2, unexplained variance) or 4-component when present (~U = 1-conf proxy) latent L = hidden-factor risk
```

EXPLANATION

Intake map footer (continuation). 5 of 6 direct edges shown; +48 edges among other scope vars; session 5 exchanges. Display metadata.

RAW LOG BLOCK — #36

```
[observer] reliability verified: intake_velocity – coverage 15/15 hops (27.2s background)

[verify] prediction intake: predicted=+104.4 actual=+106.9 error=2.5 x MISSED

[observer] New engine cycles arrived – causal model now stale (re-run DIFC to refresh)
[difc-hb] 5% Initializing... (t+0.0s)
[difc-hb] 12% Data ready: 200 cycles x 40 vars (t+0.1s)
[difc-hb] 20% EXPERIMENT mode – causal discovery (200 cycles)... (t+0.1s)
[difc-hb] 55% Physics validation... (t+6.0s)
[difc-hb] · pwe-prerefine start (t+6.0s)
[difc-hb] · pwe-prerefine done (t+19.8s)
[difc-hb] · filter-loop start (t+19.8s)
[observer] intake_velocity: scope = 9 vars (1 DAG-ancestors + 8 data-prescreened) – 36 vars is run me discovery me nahi the (har run top-correlated rotate hote)
[observer] intake_velocity: raw=12 valid=12 accepted=12 unknown=0 rejected=0 (stored=12: 12 confirmed + 0 provisional)
[difc-hb] · filter-loop done (t+19.8s)
[difc-hb] 65% Sensitivity calibration... (t+19.8s)
[difc-hb] 72% SBL learning update... (t+34.2s)
[difc-hb] 80% ACIS reasoning... (t+34.2s)
[difc-hb] 87% Querying per-target decisions... (t+34.2s)
[difc-hb] 93% Multi-objective tradeoff... (t+34.2s)
[difc-hb] 97% Building causal graph... (t+34.2s)
[difc-hb] · paths-extract start (t+34.2s)
[difc-hb] · paths-extract done (t+34.2s)
[difc-hb] · broadcast start (t+34.2s)
[difc-hb] · bcast: payload built (t+34.2s)
[difc-hb] · cdifc: entry (t+34.2s)
[difc-hb] · cdifc: edge_memory done (t+34.2s)
[difc-hb] · cdifc: freeze+carry done (t+34.3s)
[difc-hb] · cdifc: rel-verify deferred (background) (t+34.3s)
[difc-hb] · cdifc: census done (t+34.3s)
[difc-hb] · cdifc: knowledge recomputed (t+34.3s)
[difc-hb] · bcast: complete_difc done (t+34.3s)
[difc-hb] · bcast: difc_done emitted (t+34.3s)
[difc-hb] · broadcast done (t+34.3s)
[difc-hb] 100% Done – 12 edges (t+34.3s)

[observer] DIFC complete – causal structure updated
[observer] Economy: 23.0515 knowledge-value · discovery 43.237 CPU-sec · maintenance 386.385 CPU-sec
[observer] epistemic: U_total=0.25 · latent=0.68 (max; mean 0.34) (type 'ceh' for breakdown)
[ROUTER] CHAT

△ variable: ca50 DIFC scope: 18 valid edges signal=0.34 difc=VALID

drivers
Ca50 consistently increases ca90 (limited evidence, single-regime).

start investigating:
· ca50 hide karo – isolate its effect
· ascii – causal map
~ knock elevated (28.72) – check causal drivers
~ trust(ca50)=0.00 · weakest: evidence=0.00 · recover: run more experiments
~ ca50: edges stale (1020 cycles old) – re-run DIFC
```


EXPLANATION

What happened: Background verify confirms intake_velocity 15/15 (27.2s). Then the forecast is scored: **predicted +104.4, actual +106.9, error 2.5 X MISSED**. Model goes stale; a second intake run executes (12 edges, 34.3s, U_total 0.25, latent 0.68). Economy climbs to 23.05 know-value over 386.4 CPU-s maintenance. A ca50 driver chat reprints (edges now 1020 cycles stale).

What this means: A verified, low-latent, high-confidence prediction still *missed* by 2.5 units. This is the most important honest signal in the run: structural cleanliness and stated confidence did not guarantee predictive accuracy, and the system logged the miss rather than hiding it.

Evidence produced: A signed prediction error (2.5) with an explicit MISSED mark, a follow-on re-run, and rising cumulative compute cost.

Why this matters: Logging a confident miss is exactly what a trustworthy closed loop must do — the error feeds back (ORCH learning) and tempers future confidence. It is evidence the system is empirical, not self-congratulatory.

Insights: 'High confidence' here was about reversion stability, not realized accuracy — a real limitation. The miss is the kind of datum that, accumulated, should recalibrate the confidence labeling itself.

RAW LOG BLOCK — #37

```
[observer] reliability verified: intake_velocity - coverage 12/12 hops (18.7s background)
[ROUTER] COMMAND

drivers
```

EXPLANATION

Second intake verify (12/12, 18.7s) + engine stop. The re-run's smaller 12-edge graph also verifies fully; the operator then halts the engine and freezes the buffer.

RAW LOG BLOCK — #38

```
- Engine stopped
user request
```

EXPLANATION

Engine-stopped sub-banner (continuation). Logs the halt as a user request within the drivers branch — provenance only.

RAW LOG BLOCK — #39

```
Engine halted. Buffer frozen at last state.
~ knock elevated (29.13) - check causal drivers
~ ca50: edges stale (1070 cycles old) - re-run DIFC
```

EXPLANATION

Halt confirmation (continuation). Buffer frozen; knock (29.13) and ca50-staleness (1070 cycles) advisories print. ca50's graph is now very stale, underlining that the session's attention moved to intake.

RAW LOG BLOCK — #40

```
EBIS · waiting for first investigation
engine offline - 'engine chalao' se shuru karo
+ Investigation context restored: branch=drivers investigation last=ca50 vars=6 history=1
```

EXPLANATION

What happened: Session closes back to idle: 'waiting for first investigation', engine offline, and context restored — branch *drivers investigation*, last target ca50, 6 vars, history 1.

What this means: The runtime persists its final state so the next boot resumes exactly here (mirroring the opening block). The branch has evolved from 'evidence verification' to 'drivers investigation' over the session.

Evidence produced: Final restored-context fields, including a history counter (1), confirming the state was checkpointed.

Why this matters: Closing symmetry with the boot block proves end-to-end persistence — the investigation is durable across the stop, ready to continue.

Insights: The session's net movement: from verifying evidence to hunting drivers, with ca50 still the open question and intake_velocity identified as the cleanest, most predictable target found.
